

■ Scientific Justification (required for all)

Does circumnuclear dust remember an AGN’s accretion history? Large luminosity changes in active galactic nuclei (AGN) provide timed experiments on the material surrounding accreting black holes. Strongly variable quasars [1–4], including changing-look AGN [5], alter the radiation incident on circumnuclear dust over observable timescales. Existing grains heat or cool as that radiation reaches them; sufficiently large changes may also destroy grains or permit new dust to form. **Can the infrared response be explained by changing illumination of an unchanged dust distribution, including light-travel delays, or is additional evolution of the dust required?** Resolving this question is essential to interpreting infrared spectra as measures of an AGN’s luminosity and dust covering factor during variability.

We propose MIRI/MRS spectroscopy of **24 AGN, divided into 12 fading and 12 rising sources**, with SPHEREx spectra, ZTF/WISE light curves, and several optical spectra per source, including 2026 NGPS spectra. The near-infrared histories constrain the response of hot dust close to the nucleus. MIRI measures the longer-wavelength continuum and silicate features of warmer, more distant dust, which can still echo earlier illumination. Together, these observations test whether a single dust distribution accounts for both responses.

From infrared echoes to dust evolution. The equilibrium sublimation radius scales approximately as $L^{1/2}$ [6–8], but the dust distribution need not adjust immediately. A rising source tests whether increased heating alone explains the spectrum or whether the inner emitting region changes. After fading, surviving grains cool, while dust formation at smaller radii may eventually restore a hotter component. Recovery after a decline provides a further test of whether the spectrum depends on the path to the present luminosity.

WISE has established infrared echoes of changing-look transitions [9–12] and enabled variability-based searches for these sources [10,12,13]. Broadband colours alone, however, cannot separate changes in temperature, emitting area, extinction and host dilution [14]. Reverberation studies illustrate the physical alternatives: the inner dust radius in Mrk 590 decreased after a luminosity decline, suggesting replenishment within a few years [15], whereas the hottest dust in NGC 4151 remained at approximately the same radius over decades, with a much longer response from the warm component [16]. Recent MRS observations reveal spectral diversity beyond current dust models [17], including the distinctive silicate spectrum of the recently activated nucleus Ansky [18]. Connecting that diversity to measured illumination histories is the next step.

Why MIRI? SPHEREx covers 0.75–5 μm [19], tracing the hot-dust spectrum. At longer wavelengths the variability is damped and delayed [20], so the warm emission records a different portion of the preceding history. MIRI/MRS extends coverage to 27.9 μm and resolves the warm continuum, silicate features, aromatic bands and ionic lines [21]. Its spatial information also separates the nuclear point source from measurable extended host emission. These constraints are necessary to distinguish a delayed thermal response from a change in the emitting dust.

A sample selected by variability history. All targets have positions in a low-dimensional manifold of ZTF/WISE variability. Their measured histories define the rising and fading comparison groups, including catalogue CLAGN [22–25]; broad-line appearance or disappearance is not required. The archival optical record is extended with 2026 NGPS spectra to constrain

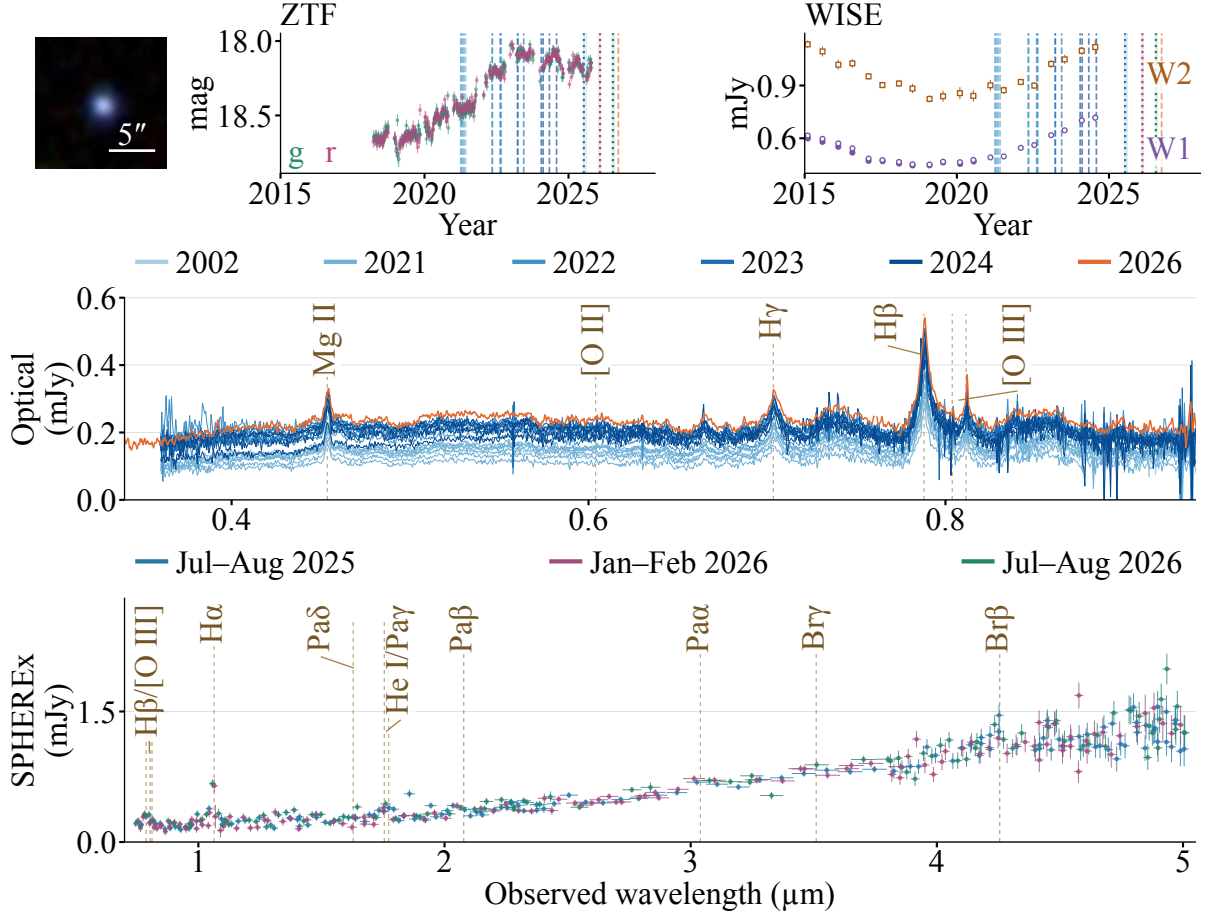


Figure 1: **P1823 illustrates the measured baseline for the dust-response experiment.** At $z = 0.620$, the quasar is in a high optical state following an earlier WISE decline and recovery. Top: SDSS image (5'' bar), ZTF g, r , and WISE W1/W2 light curves. Middle: 21 optical spectra. Bottom: three SPHEREx visits. Matching colours connect the spectra to their dates on the light curves; shading spans each SPHEREx visit. The 2002 optical epoch predates the displayed window. Spectra use observed wavelength and mJy; gold markers indicate expected line positions.

continuum and broad-line changes. Figure 1 shows the available sequence for P1823: a sustained optical rise, an infrared decline followed by recovery, and three dated SPHEREx visits. P11530 provides a contrasting history, fading over 2018–2021 and partly recovering in 2022–2023.

The relevant comparison is the rest-frame age of a luminosity change, $\Delta t = (t_{\text{JWST}} - t_{\text{change}})/(1+z)$, relative to the hot- and warm-dust delays, τ_{dust} and τ_{warm} . The delays depend on luminosity [8,20,26]. We will bracket changes with optical data and estimate delays from luminosities or reverberation constraints, propagating event-date, host and lag uncertainties. Earlier reversals remain in each source’s illumination history.

Testing fixed and evolving dust. We will first fit the optical history, WISE photometry and SPHEREx spectra with a fixed dust distribution, including light-travel delays, extinction and host emission. Clumpy and disk-plus-wind geometries [31–33] provide alternative descriptions of the unresolved dust. The fitted model predicts the warm continuum and silicate spectrum at the MIRI epoch. We will then allow the amount or location of inner dust to evolve, retaining the same treatment of illumination and observational uncertainties. Agreement with fixed dust constrains structural changes; a history-dependent discrepancy that persists across geometry and host models would support dust evolution.

Our primary diagnostic is

$$\delta_{\text{warm}} = \log_{10} \left(\frac{L_{8-13\mu\text{m}}^{\text{MRS}}}{L_{8-13\mu\text{m}}^{\text{fixed}}(t_{\text{MIRI}})} \right),$$

the residual nuclear warm-dust continuum luminosity relative to the fixed-distribution prediction. We will compare this residual between the two groups and against continuous $\Delta t/\tau_{\text{dust}}$ and $\Delta t/\tau_{\text{warm}}$, accounting for luminosity and host contribution. As a secondary diagnostic, the rest 2–5 to 8–13 μm luminosity ratio connects the model-interpolated hot component to the MIRI measurement.

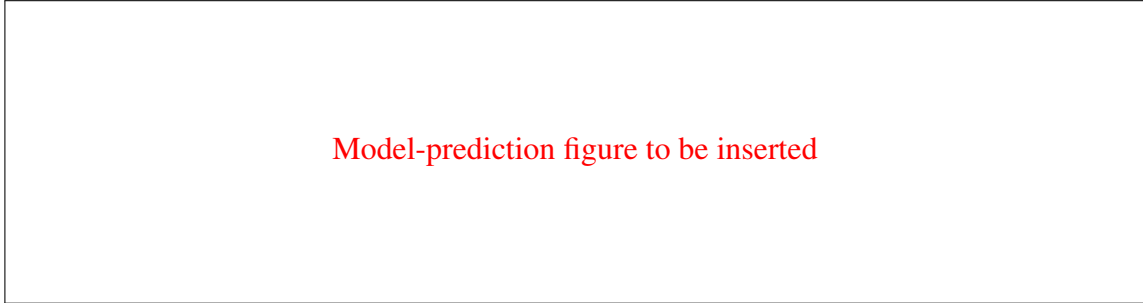


Figure 2: Fixed and evolving dust predictions for δ_{warm} as a function of event age relative to dust delay, with target positions and expected measurement uncertainties. Quantitative predictions remain to be calculated.

Why 24 sources? Twelve objects per group provide replication across different luminosity histories and reduce sensitivity to individual peculiarities. For independent measurements with a common within-group scatter s , the uncertainty on each mean is $s/\sqrt{12} = 0.289s$ and on the rising–fading difference is $s\sqrt{2/12} = 0.408s$. The expected scatter of **X dex** and model separation of **Y dex** set the quantitative sensitivity of the experiment (Figure 2). The inference will include unequal errors, intrinsic scatter and shared calibration and model uncertainties; removing individual sources will test robustness. The result concerns dust response in this deliberately selected sample, rather than a population-wide transition rate.

■ Description of Observations (required for all GO and Survey)

Spectral coverage and sensitivity. We request one MIRI/MRS visit per target, using all three grating settings, target acquisition and a four-point dither to cover approximately $4.9\text{--}27.9\ \mu\text{m}$ [21,28]. Twenty-three targets have $z < 0.3$, placing both silicate peaks within MRS. P1823 provides the well-sampled optical/SPHEREx baseline at higher redshift: its $9.7\ \mu\text{m}$ feature lies at $15.7\ \mu\text{m}$, while its $18\ \mu\text{m}$ peak is outside MRS. The primary diagnostic is accessible for every target and does not depend on channel 4C.

Archival total-source W3 and W4 fluxes span approximately $1.9\text{--}48.4$ and $3.7\text{--}82.6$ mJy. Exposure design will use conservative nuclear continua and host backgrounds, with a target S/N of **50** per $R = 100$ bin near rest 5 and $12\ \mu\text{m}$ and **30** across the accessible $18\ \mu\text{m}$ feature. The readout is **FASTR1**; dedicated sky observations accompany the **M** targets requiring separate host/background measurements, while suitable compact sources use dither-pair subtraction [28]. The total request, including overheads, is **X h** for a Medium program [27].

Connecting the observing epochs. WISE and SPHEREx data enter the model at their actual dates. We will propagate uncertainty in the illumination between those measurements and the MIRI visit, rather than treating archival spectra as contemporaneous. Optical monitoring and spectroscopy near the JWST epoch will constrain intervening variability. The comparison uses the full preceding history and requires one MIRI epoch per source, without a rapid-response trigger or coordinated near-infrared visit.

Separating nuclear dust from the host. We will decompose each MRS cube into a point source and extended emission [17], then fit the continuum, silicates, aromatic bands, H_2 and ionic lines [30]. Aromatic bands and $[\text{Ne II}]$ constrain host contamination; $[\text{Ne V}]$ and $[\text{Ne VI}]$ measurements or limits constrain the hard ionizing component. Optical spectral decomposition will calibrate the photometric histories to nuclear continuum and broad-line luminosities. Unobserved intervals and the conversion to dust-heating luminosity enter as uncertainties.

The 2010 W3/W4 measurements constrain earlier emission in larger apertures. We will predict these fluxes by integrating redshifted nuclear-plus-host models through the WISE bandpasses, including colour corrections and uncertain emission outside MRS coverage. Archival imaging constrains host light beyond the MRS field and informs the SPHEREx aperture model. Where the nuclear baseline cannot be isolated, it will enter as an interval or upper bound. Wavelength-dependent calibration covariance [29] and alternative host decompositions will be included in the model comparison. Residuals common to both history groups would point to limitations of the model family; a reproducible dependence on illumination history would provide the stronger evidence for dust evolution.

Supplemental Information

Special requirements. No timing constraints, coordinated parallels or joint-facility time are requested. Dedicated MRS sky observations will use the applicable background-observation requirements [28]. [\[Target-specific sky requirements and APT visibility to be finalized.\]](#)

Duplications. [\[Target-by-target archive and approved-program duplication results.\]](#)

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